Answers

Exam in Introduction to economics for sport management

Instructions (read carefully)

a. **You can use calculators.**
b. **Justify all your answers using economic concepts, theories and reasoning.**

Good Luck!

Question 1 (20 points)

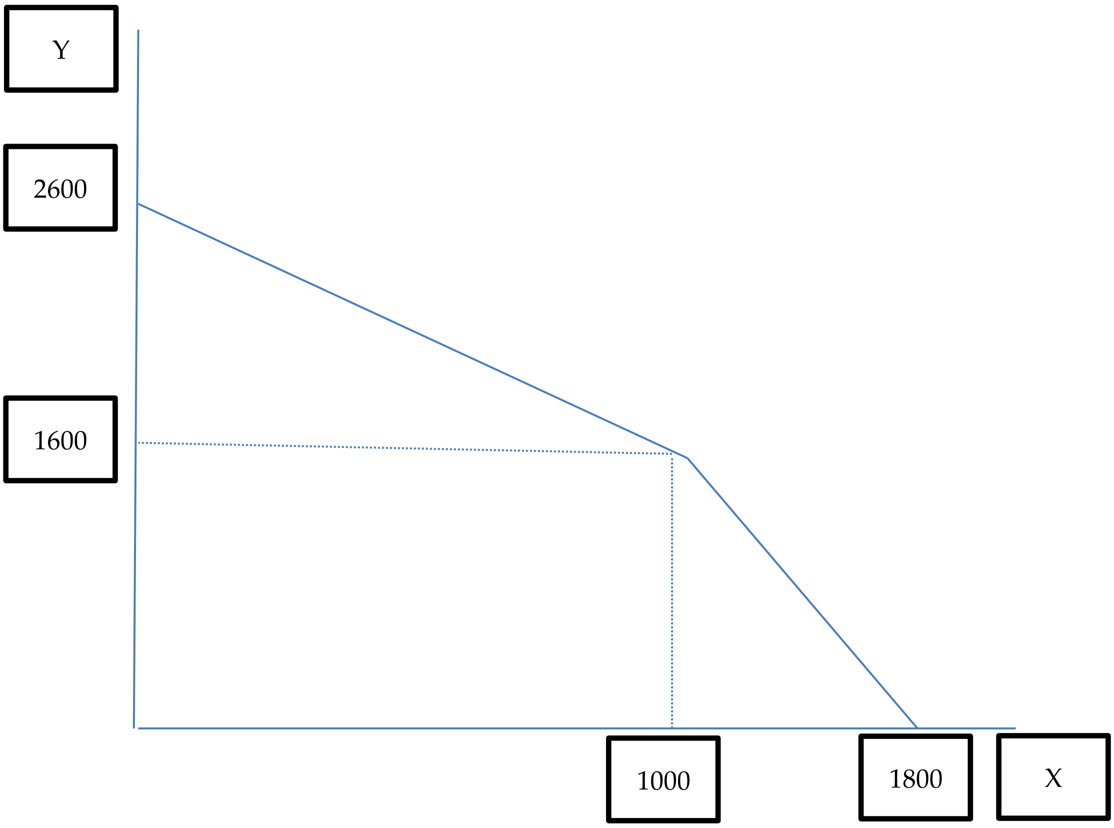
Suppose there is a country that can produce two goods: X and Y. The country has 400 machines and 200 workers. Each machine can produce 2 units of X or 4 units of Y. Each worker can produce 5 units of X or 5 units of Y.

a. Please draw a production possibilities frontier of this country and mark all the points (X and Y) where the curve changes its slope as well as the maximal production of X and Y. **(8 points)**
b. Suppose that this country needs to consume 1000 units of X. In that case how many units of Y will it consume? **(3 points)**
c. Suppose, there is a world trade, such that 1 unit of X is worth 2.5 units of Y. How many units of X and Y should this country produce and why? **(4 points)**
d. Suppose that during the world trade, this country still needs to consume 1000 units of X. In that case how many units of Y will it consume? **(5 points)**

Answers:

	machines	workers
Production X per one unit	2	5
Production Y per one unit	4	5
MC in producing X in terms of Y	2	**1**
MC in producing Y in terms of X	**0.5**	1
Max X	400*2=800	200*5=1000
Max Y	400*4=1600	200*5=1000

a. See answer a below.
b. The country will consume 1600 units of Y as shown on the graph below.
c. This country has a comparative advantage in producing X, because in the worst case, its cost of producing X is 2Y. However they can sell it to the world for the price of 1X=2.5Y. Therefore, this country will only produce 1800X and zero Y.
d. If a country needs to consume 1000 units of X, then it will still produce 1800 units of X. The country will exchange the remaining 800 units of X for 800*2.5=2000 units of Y.



Question 2 (20 points)

Suppose the following demand function: Q^D=16-2p. In addition suppose that there are <u>two</u> producers. Each producer's supply function is: $p = q^S$ where Q^D is the quantity demanded of a good, q^S is the supplied quantity of a <u>single</u> supplier of a good and p is the price of a good in USD.

a. What is the aggregate supply function in that market? **(5 points)**
b. What would be the price and the quantity in the equilibrium? **(4 points)**
c. Show on the graph (where appropriate) the cases where p=0, Q=0, and price and quantity in equilibrium. **(8 points)**

d. What is the total social welfare? **(3 points)**

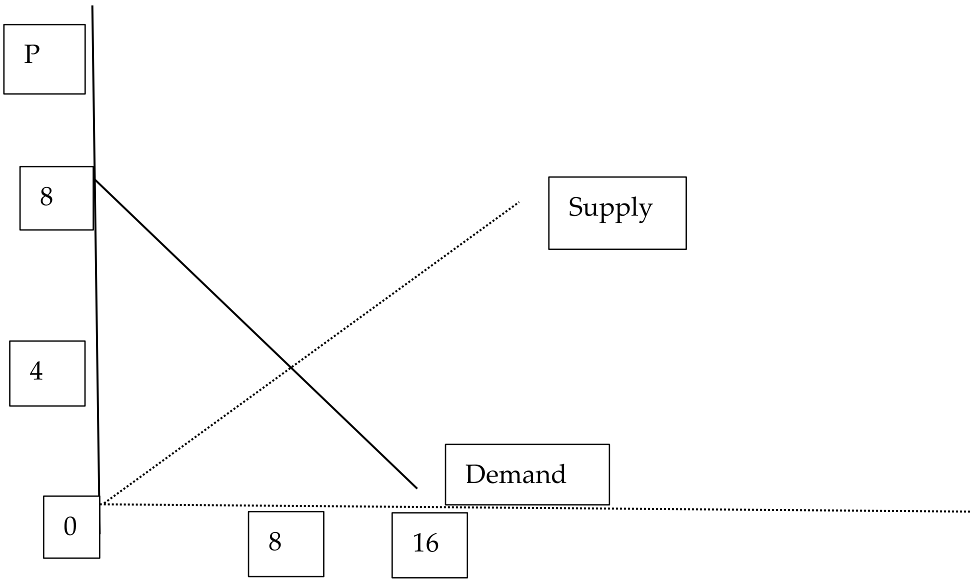
Answer:

a. **The aggregate supply function is the sum of the quantities of each of the two suppliers, namely** $Q^S = p + p = 2p$

b. **In equilibrium the supply is equal to demand (**$\boldsymbol{Q^S = Q^D}$**) therefore, 16-2p=2p. Solving this equation we get p=4 and Q=8.**

c. **See the answer below**

d The total social welfare is the sum of consumer and producer surplus.
The consumer surplus is the area between the demand curve and the price (triangle area) (8-4)*(8-0)/2=16
The producer surplus is the area between the supply curve and the price (triangle area) (4-0)*(8-0)/2=16
The total social welfare is 16+16=32



Question 3 (20 points)

There is a market with 5 workers who should be allocated between two fields with the following production function:

Field A

L	TP
0	0
1	7
2	13
3	16
4	18
5	19

Field B

L	TP
0	0
1	4
2	7
3	9
4	10
5	10

a. How many workers would work on each field if it is known that the price of one produced unit on field A is 5USD and the price of one produced unit on field B is 1USD? **(6 points)**
b. What would be the salary of each worker? **(6 points)**
c. What would happen to the profit of the owner of field A if there will be 6 workers? **(8 points)**

Answer:

Field A

L	TP	mp	Vmp (p*mp)
0	0		
1	7	7	35
2	13	6	30
3	16	3	15
4	18	2	10
5	19	1	5

Field B

L	TP	mp	Vmp (p*mp)
0	0		
1	4	4	4
2	7	3	3
3	9	2	2
4	10	1	1
5	10	0	0

a. There will be 5 workers in Field A and zero workers in field B.
b. The value of the marginal product of the last worker is 5 and this will be the wage as well (VMP=W).

Additional worker will reduce the wage to 4, but the productivity of field A will remain the same. That means that the profit of the owner of field A will be increased.

Question 4 (15 points)

Assume a firm with the following production function:

TP (Q)	L	MP	FC	VC	TC	MC	ATC	AVC
0	0				20			
1	8							
2	15							
3	27							
4	40							

It is known that one unit of labor (L) costs 20 USD.

a. Complete the table. **(8 points)**
b. What is the minimal price that the firm will produce in the short term? **(1 points)**
c. What is the minimal price that the firm will produce in the long term? **(1 points)**
d. Would other firms enter this market if the price is 160? Please explain your answer. **(5 points).**

Answer:

TP (Q)	**L**	**MP**	**FC**	**VC**	**TC**	**MC**	**ATC**	**AVC**
0	**0**		**20**		**20**			
1	**8**	**1/8**	20	**160**	**180**	**160**	**180**	**160**
2	**15**	**1/7**	20	**300**	**320**	**140**	**160**	**150**
3	**27**	**1/12**	20	**540**	**560**	**240**	**186.7**	**180**
4	**40**	**1/13**	20	**800**	**820**	**260**	**205**	**200**

b. min AVC=150. Below that price the firm will not produce in the short term
c. min ATC=160. Below that price the firm will not produce in the long term
d. When the price is 160, the long term profit is zero, therefore new firms will not enter the market, because if they enter the market, the supply becomes larger and the price will be below minATC.

Question 5 (10 points)

You were offered to invest 1200 NOK in one of the following projects:

1. Factory that yields 500 NOK at the end of the second year, 600 NOK at the end of the fourth year, and 200 NOK at the end of the fifth year. The annual interest rate is 2% in the first three years and 5% after that.
2. Football club that yields 50 NOK at the end of the first year, 100 NOK at the end of the second year, and then 500 NOK at the end of each of the following 3 years. The annual interest rate is 12%.

What would be your best investment strategy and why.

Answer:

1. **$PV=500/((1.02)^2)+600/((1.02)^3*(1.05))+200//((1.02)^3*(1.05)^2)=1189$**
2. **$PV=50/(1.12)+100/(1.12)^2+500/(1.12)^3+/(1.12)^4+/(1.12)^5=1081$**

Option 1 is the best option since it yields the highest PV, however it is not worth to invest since its PV is lower than the investment (1200).

Question 6 (10 points)

Suppose a good X with demand elasticity of 1. Suppose, there is a technological change that allows suppliers to produce 10% more for the same cost. What would happen to suppliers' total revenue compared to the case before the technological change?

Answer:

As a result of technological change the supply will move down and to the right (an increased supply), however the demand curve will remain the same. The price will be decreased, however the total revenue will remain the same since the demand elasticity is 1.

Question 7 (5 points)

One of the possible gains in hosting Mega-Events is an increased number of tourists after the event. Please discuss the example of the 2010 FIFA World Cup on whether tourism from non-SADC countries has increased and whether the cost of stadium constructions was justified in terms of tourists' spending during the event.

Answer from the presentation:

The 2010 FIFA World Cup entailed $3.9 billion in expenses borne by South Africa ($1.3 billion in stadium construction costs) (Du Plessis and Maenning, 2011, Development Southern Africa).

Even when the authors take a very conservative view and only consider the cost of stadium construction, the estimates still point to a cost of over $4,400 per additional non-SADC visitor.

Also, there was no substantial growth in tourism from non-SADC countries after the event.